Variables

Variables

A core of programming

Variables hold *variable* value

In the context of *Computer Science* variables are a **name** with an associated **value**

Patterns with variables

We'll look at some common ways to use variables in Scratch

Each shows how variables hold values and change them over time

The goal is

- to develop an intuitive understanding of how variables work
- to understand which situations variables will be helpful in
- to recall common patterns that are useful for a large majority of programs

Pattern 1: Configuration Bundle

The problem: How do you make the game *feel* right?

Player movement uses multiple connected values:

- `speed` – how fast you move sideways
- `jumpForce` – how high you go
- `gravity` – how fast you come back down

These variables form a **system** – change one, and the whole feel of the game changes.

Pattern 1: Configuration Bundle

```
event
when [flag v] clicked
variables
set [speed v] to [5]
variables
set [jumpForce v] to [12]
variables
set [gravity v] to [1]


control
forever
  motion
  change x by [speed]
  change y by [jumpForce]
  change y by [-gravity]

```

These variables are used **inside other blocks** – they control how the sprite moves.

Pattern 1: Configuration Bundle

Different preset combinations produce completely different game feel:

speed	jumpForce	gravity	Feel
3	8	0.5	Floaty, slow moon-jump
5	12	1	Standard platforming
8	15	2	Fast, heavy

The variables work as a **coordinated system** — real game developers tune these together to get the right feel.

Pattern 1: Configuration Bundle

Variables aren't just for tracking – they're **settings** you can tweak.

This bundle pattern is especially useful because:

- All tuning values are in **one place** (not scattered across blocks)
- You can **experiment** quickly by changing one number
- Other developers can understand your game's feel at a glance

Pattern 1: Configuration Bundle

We'll add speed, jumpForce, and gravity to our game.

Find a combination that feels fun to play with.

Pattern 2: Counter (Score)

The problem: How would you count how many coins a player collects?

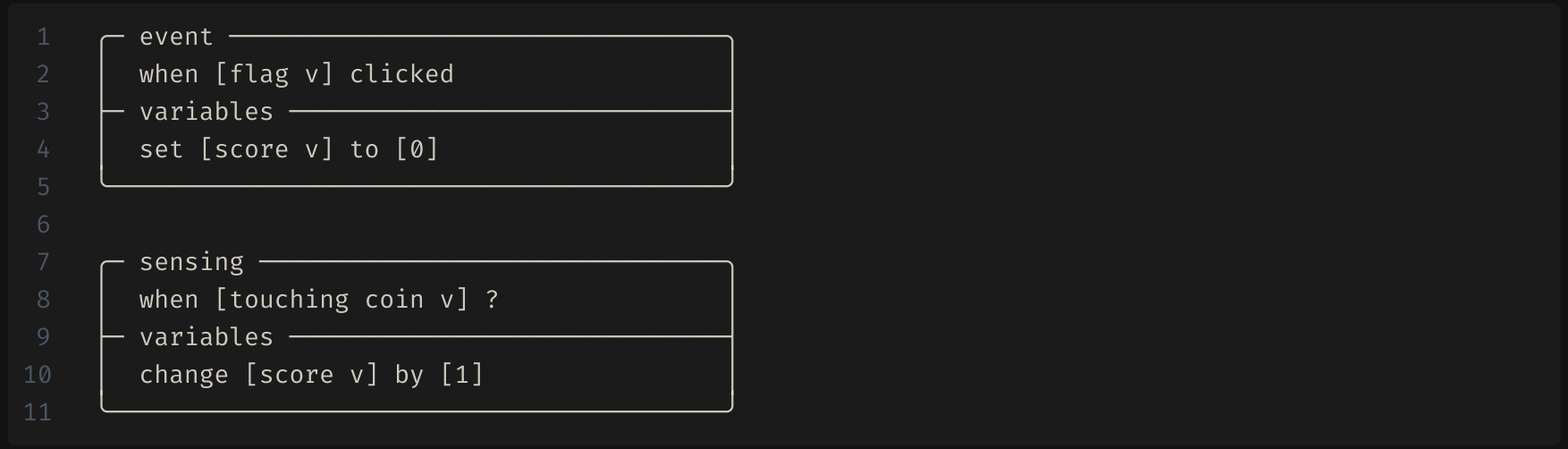
Every time the player touches a coin, the score goes up by 1.

> Turn to a partner: describe in plain English what the program needs to do

We need:

1. A place to store the count → a **variable**
2. A starting value → **0**
3. Something that triggers the increase → **touching a coin**
4. The change itself → **+1**

Pattern 2: Counter (Score)



Two separate stacks:

- one to *initialize*,
- one to *respond* to an event.

Pattern 2: Counter (Score)

Step	Event	`score` value
Start	Game starts	0
1	Touch coin 1	1
2	Touch coin 2	2
3	Touch coin 3	3
…	…	…

Each event → value goes up by exactly 1.

The variable **remembers** the total so far between events.

Pattern 2: Counter (Score)

We'll make a game where every time the player collects an item, the score increases by 1.

We'll also make different items worth different amounts.

Pattern 3: Countdown (Timer)

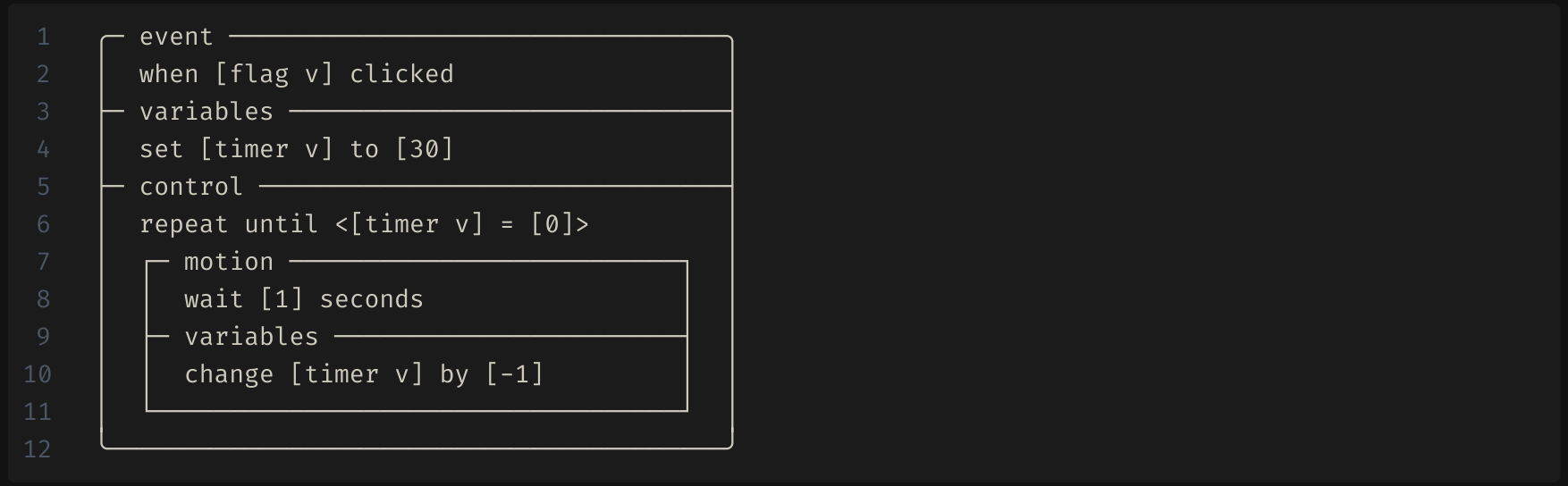
The problem: How do you make a timer that counts down from 30?

When the timer reaches 0, the game ends.

> How is this different from counting up?

The change is **negative** instead of positive, and you **stop** when you reach a target value.

Pattern 3: Countdown (Timer)



negative change, **loop**, and **stop condition**.

Pattern 3: Countdown (Timer)

Step	Action	`timer` value
Start	Initialize	30
1	Wait 1 sec, change by -1	29
2	Wait 1 sec, change by -1	28
…	…	…
30	Wait 1 sec, change by -1	**0 → stop**

The variable **drives the loop** – the loop checks the value each time and stops when it hits 0.

Pattern 3: Countdown (Timer)

To the same game, we'll make it end after 15 seconds.

And then change how the player looks when there's only 5 seconds left.

Pattern 4: Flag (State Variable)

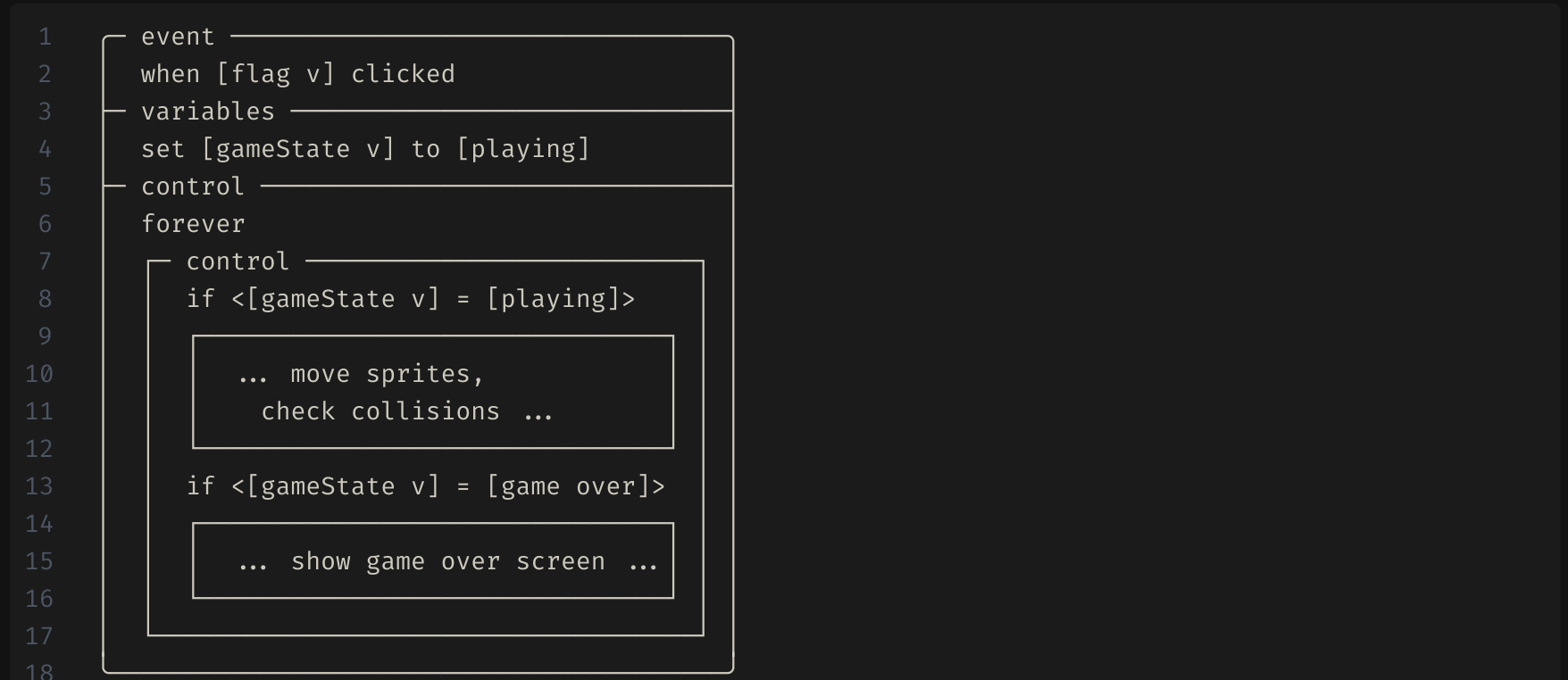
The problem: How does the game know whether it's playing, paused, or over?

You need a variable that stores the **current state** of the game.

Common states:

- `"playing"`,
- `"paused"`,
- `"game over"`,
- `"menu"`

Pattern 4: Flag (State Variable)



Pattern 4: Flag (State Variable)

A flag variable doesn't count – it **switches** between values:

```
gameState:  menu  →  playing  →  game over
                  ↑          ↓
                  └ paused ──┘
```

The variable acts as a **traffic light** – code checks it before deciding what to do.

This replaces a tangled mess of `if` blocks with a single variable that controls the flow.

Pattern 4: Flag (State Variable)

We'll add 2 states to our game:

- playing
- game over

And 2 states for the player:

- moving
- idle

Pattern 5: Max Tracker (High Score)

The problem: How do you remember the highest score ever achieved?

Every time the score changes, check if it's higher than the previous best.

> What variable do we need beyond the current score?

A second variable: `highScore`. It only changes when `score` exceeds it.

Pattern 5: Max Tracker (High Score)

```
event
when [flag v] clicked
variables
set [highScore v] to [0]


event
when [score changes]
control
if <[score v] > [highScore v]>
 variables
 set [highScore v] to [score]
```

A **conditional update** – the variable only changes when a comparison is true.

Pattern 5: Max Tracker (High Score)

Round	Score after round	Comparison	`highScore`
1	15	15 > 0 → true	15
2	12	12 > 15 → false	15
3	22	22 > 15 → true	**22**
4	18	18 > 22 → false	22
5	30	30 > 22 → true	**30**

`highScore` only moves **upward**. It holds the "best so far."

This pattern generalizes to: **compare → conditionally update** (works for min, max, closest, etc.)

Pattern 5: Max Tracker (High Score)

We'll add a high score to our game. If we click the flag, the score resets but the high score stays until we close the browser.

Summary: What all 5 patterns share

Each pattern is the same core idea:

> A **named box** (variable) whose **contents change** over time.

The pattern is defined by *what* changes the value and *when*.

Pattern	What changes the value	When
Bundle	No change (it *drives*)	At start, then read only
Counter	Event	Each time event occurs
Countdown	Loop iteration	Each second until 0
Flag	Game event	When state transitions
Max Tracker	Another variable	When it exceeds current